

Review of Seeber's Investigations

English translation of the identity passage from Gauss, Werke, Band II, p. 193

C. F. Gauss

Translation prepared from the source-locked TeX retype in this packet

Preliminary note

Let the positive ternary quadratic form

$$axx + byy + czz + 2dyz + 2bzx + 2cxy$$

(whose determinant is $= -D$) be reduced in Seeber's sense; by a', b', c' are denoted those coefficients which occur in the preliminary equations (...) as $2a', 2b', 2c'$.

I. Case: none of the numbers a', b', c' is negative

Put

$$\begin{aligned} b - 2a' &= d, & c - 2b' &= e, & a - 2c' &= f, \\ c + 2a' &= g, & a + 2b' &= h, & b + 2c' &= i, \end{aligned}$$

where, from the definition of reduced positive forms, it follows at once that if $axx + byy + czz + 2dyz + 2bzx + 2cxy$ is such a form, none of those six numbers is negative. Now if the (negative) determinant of the form is denoted by $-D$, then, as is easily verified by expansion, one has the identity

$$2D - abc = a a' d + b b' e + c c' f + a' h i + b' g i + c' g h + g h i$$

in which none of the seven terms on the right can be negative, and consequently abc is not greater than $2D$. The same follows in the same way from the identity

$$2D - abc = a a' g + b b' h + c c' i + a' e f + b' d f + c' d e + d e f$$

II. Case: none of the numbers a', b', c' is positive

Put

$$\begin{aligned} b + 2a' &= d, & c + 2b' &= e, & a + 2c' &= f, \\ c + 2a' &= g, & a + 2b' &= h, & b + 2c' &= i, \\ b + c + 2a' + 2b' + 2c' &= k, \\ a + c + 2a' + 2b' + 2c' &= l, \\ a + b + 2a' + 2b' + 2c' &= m, \end{aligned}$$

and let the determinant of the form be, as before, $= -D$. By virtue of the definition of reduced positive forms, none of the nine numbers $d, e, f, g, h, i, k, l, m$ can be negative; and so from the identity

$$6D - 3abc = -a a'(d + 2k) - b b'(e + 2l) - c c'(f + 2m) \\ - a' h i - b' g i - c' g h + d e f + 2 g h i$$

all terms on the right become positive or zero, since a', b', c' are not positive but negative or zero. Thus $3abc$ cannot be greater than $6D$, or abc cannot be greater than $2D$. The same likewise follows from the identity

$$6D - 3abc = -a a'(g + 2k) - b b'(h + 2l) - c c'(i + 2m) \\ - a' e f - b' d f - c' d e + 2 d e f + g h i.$$

Simplified unsymmetric identity

Both identities are symmetric. If one gives up complete symmetry, the proof can be carried out with an even smaller number of terms, for example by the identity

$$8D - 4abc = -2 a a'(g + k) - 2 b b'(e + l) - 4 c c' m + (c + e) d f + (c + g) h i$$

Translation note: the algebraic notation is intentionally kept parallel to the German source; the determinant sign convention $= -D$ and the primed coefficients a', b', c' are preserved without normalization.